

29.1

Cmm Model (1)

29.1.1 Cmm Model: ISRO CSE 2017 | Question: 47



Capability maturity Model (CMM) is the methodology to

- A. develop and refine an organization's software development process
- B. develop the software
- C. test the software
- D. All of the above

isro2017 is&software-engineering non-gatecse cmm-model

Answer key

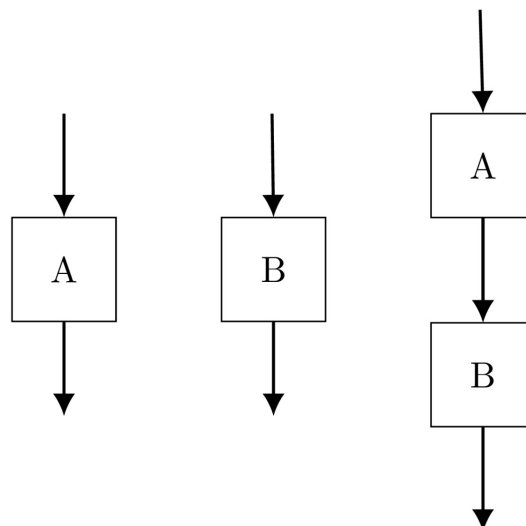
29.2

Cyclomatic Complexity (6)

29.2.1 Cyclomatic Complexity: GATE CSE 2010 | Question: 21,ISRO2017-60



The cyclomatic complexity of each of the modules A and B shown below is 10. What is the cyclomatic complexity of the sequential integration shown on the right hand side?



- A. 19
- B. 21
- C. 20
- D. 10

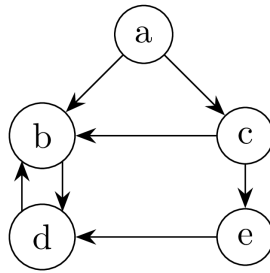
gatecse-2010 is&software-engineering cyclomatic-complexity easy isro2017

Answer key

29.2.2 Cyclomatic Complexity: ISRO CSE 2011 | Question: 65



The Cyclomatic Complexity metric $V(G)$ of the following control flow graph



- A. 3 B. 4 C. 5 D. 6

isro2011 is&software-engineering cyclomatic-complexity

Answer key

29.2.3 Cyclomatic Complexity: ISRO CSE 2014 | Question: 4



Consider the following pseudo- code

```

while (m<n)
  if (x>y ) and (a<b) then
    a=a+1
    y=y-1
  end if
m=m+1 end while
  
```

What is cyclomatic complexity of the above pseudo -code?

- A. 2 B. 3 C. 4 D. 5

isro2014 is&software-engineering cyclomatic-complexity non-gatecse

Answer key

29.2.4 Cyclomatic Complexity: ISRO CSE 2020 | Question: 69



Consider the following pseudo-code

```

I=0; J=0; K=8;
while(I<K-1) //while-1
{
  J=J+1;
  while(J<K) //while-2
  {
    if(x[I]<x[J])
    {
      temp = x[I];
      x[I]=x[J];
      x[J]=temp;
    }
  }
  I=I+1;
} // end of while-1
  
```

The cyclomatic complexity of the above is

- A. 3 B. 2 C. 4 D. 1

isro-2020 is&software-engineering cyclomatic-complexity normal

Answer key

29.2.5 Cyclomatic Complexity: ISRO CSE 2023 | Question: 61



McCabe's cyclomatic complexity number of a program control graph 'G' with e edges, n nodes and p disconnected paths is defined as

- A. $n - e + 2p$
- B. $e - n + 2 + p$
- C. $e - n + 2p$
- D. $n - e + p$

isro-cse-2023 is&software-engineering non-gatecse cyclomatic-complexity

Answer key

29.2.6 Cyclomatic Complexity: ISRO-2013-66



What is the cyclomatic complexity of a module which has seventeen edges and thirteen nodes?

- A. 4
- B. 5
- C. 6
- D. 7

isro2013 is&software-engineering cyclomatic-complexity non-gatecse

Answer key

29.3 Is&software Engineering (23)

29.3.1 Is&software Engineering: ISRO 2017-39



Estimation at software development effort for organic software in basic COCOMO is:

- A. $E = 2.0 \text{ (KLOC)}^{1.05} \text{ PM}$
- B. $E = 3.4 \text{ (KLOC)}^{1.06} \text{ PM}$
- C. $E = 2.4 \text{ (KLOC)}^{1.05} \text{ PM}$
- D. $E = 2.4 \text{ (KLOC)}^{1.07} \text{ PM}$

isro2017 is&software-engineering non-gatecse

Answer key

29.3.2 Is&software Engineering: ISRO CSE 2007 | Question: 28



The level of aggregation of information required for operational control is

- A. Detailed
- B. Aggregate
- C. Qualitative
- D. None of the above

isro2007 non-gatecse is&software-engineering

Answer key

29.3.3 Is&software Engineering: ISRO CSE 2007 | Question: 76



Which of the following is not an approach to Software Process Assessment?

- A. SPICE(ISO/IEC15504)
- B. Standard CMMI Assessment Method for process improvement
- C. ISO 9001:2000
- D. IEEE 2000:2001

isro2007 is&software-engineering

Answer key

29.3.4 Is&software Engineering: ISRO CSE 2007 | Question: 77



A physical DFD specifies

- A. what processes will be used
- B. who generates data and who processes it
- C. what each person in an organization does
- D. which data will be generated

isro2007 is&software-engineering

Answer key

29.3.5 Is&software Engineering: ISRO CSE 2011 | Question: 17



A context model of a software system can be shown by drawing a

- A. LEVEL-0 DFD B. LEVEL-1 DFD C. LEVEL-2 DFD D. LEVEL-3 DFD

isro2011 is&software-engineering

Answer key

29.3.6 Is&software Engineering: ISRO CSE 2011 | Question: 51



Warnier Diagram enables the analyst to represent

- A. Class structure B. Information Hierarchy
C. Data Flow D. State Transition

isro2011 is&software-engineering non-gatecse

Answer key

29.3.7 Is&software Engineering: ISRO CSE 2014 | Question: 20



Which of the following is not a maturity level as per Capability Maturity Model?

- A. Initial B. Measurable C. Repeatable D. Optimized

is&software-engineering non-gatecse isro2014

Answer key

29.3.8 Is&software Engineering: ISRO CSE 2014 | Question: 69



For a software project, the spiral model was employed. When will the spiral stop?

- A. When the software product is retired
B. When the software product is released after Beta testing
C. When the risk analysis is completed
D. After completing five loops

isro2014 is&software-engineering non-gatecse

Answer key

29.3.9 Is&software Engineering: ISRO CSE 2016 | Question: 61



The extent to which the s/w can continue to operate correctly despite the introduction of invalid inputs is called as

- A. Reliability B. Robustness C. Fault tolerance D. Portability

is&software-engineering isro2016

Answer key

29.3.10 Is&software Engineering: ISRO CSE 2016 | Question: 62



Which one of the following is a functional requirement ?

- A. Maintainability B. Portability
C. Robustness D. None of the mentioned

is&software-engineering isro2016

Answer key

29.3.11 Is&software Engineering: ISRO CSE 2016 | Question: 63



Configuration management is not concerned with

- A. controlling changes to the source code
- B. choice of hardware configuration for an application
- C. controlling documentation changes
- D. maintaining versions of software

non-gatecse is&software-engineering isro2016

Answer key

29.3.12 Is&software Engineering: ISRO CSE 2016 | Question: 65



A company needs to develop a digital signal processing software for one of its newest inventions. The software is expected to have 20000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.2 for the software development on embedded systems, while the exponentiation factor is given as 1.5. What is the estimated effort in person-months?

- A. 196.77
- B. 206.56
- C. 199.56
- D. 210.68

non-gatecse is&software-engineering isro2016

Answer key

29.3.13 Is&software Engineering: ISRO CSE 2017 | Question: 58



Which of the following is not a life cycle model?

- A. Spiral model
- B. Prototyping model
- C. Waterfall model
- D. Capability maturity model

isro2017 is&software-engineering non-gatecse

Answer key

29.3.14 Is&software Engineering: ISRO CSE 2017 | Question: 61



In software maintenance tackling the changes in the hardware or software environment where the software works, is

- A. Corrective maintenance
- B. Perfective maintenance
- C. Adaptive maintenance
- D. Preventive maintenance

isro2017 is&software-engineering non-gatecse

Answer key

29.3.15 Is&software Engineering: ISRO CSE 2017 | Question: 65



Which product metric gives the measure of the average length of words and sentence in documents?

- A. SCI number
- B. Cyclomatic complexity
- C. LOC
- D. Fog index

isro2017 is&software-engineering non-gatecse

Answer key

29.3.16 Is&software Engineering: ISRO CSE 2018 | Question: 47



In a particular program, it is found that 1% of the code account for 50% of the execution time. To code a program in C++, it takes 100 man-days. Coding in assembly language is 10 times harder than coding in C++, but runs 5 times faster. Converting an existing C++ program into an assembly language program is 4 times faster.

To completely write the program in C++ and rewrite the 1% code in assembly language, if a project team needs 13 days, the team consist of

- a. 13 programmers
- b. 10 programmers
- c. 8 programmers
- d. 100/13 programmers

isro2018 is&software-engineering

Answer key

29.3.17 Is&software Engineering: ISRO CSE 2018 | Question: 59



The lower degree of cohesion is kind of:

- a. Logical Cohesion
- b. Coincidental Cohesion
- c. Procedural Cohesion
- d. Communicational Cohesion

isro2018 is&software-engineering

Answer key

29.3.18 Is&software Engineering: ISRO CSE 2018 | Question: 76



Cloaking is a search engine optimization (*SEO*) technique. During cloaking:

- a. Content presented to search engine spider is different from that presented to user's browser
- b. Content present to search engine spider and browser is same
- c. Contents of user's requested website are changed
- d. None of the above

isro2018 is&software-engineering

Answer key

29.3.19 Is&software Engineering: ISRO CSE 2020 | Question: 64



Regression testing is primarily related to

- A. Functional testing
- B. Development testing
- C. Data flow testing
- D. Maintenance testing

isro-2020 is&software-engineering normal

Answer key

29.3.20 Is&software Engineering: ISRO CSE 2020 | Question: 67



What is the availability of the software with the following reliability figures

Mean Time Between Failures (MTBF) is 20 days

Mean Time To Repair (MTTR) is 20 hours

- A. 90%
- B. 96%
- C. 24%
- D. 50%

isro-2020 is&software-engineering normal

Answer key

29.3.21 Is&software Engineering: ISRO CSE 2020 | Question: 68



What is the defect rate for Six sigma?

- A. 1.0 defect per million lines of code
- B. 1.4 defects per million lines of code
- C. 3.0 defects per million lines of code
- D. 3.4 defects per million lines of code

isro-2020 is&software-engineering normal

Answer key

29.3.22 Is&software Engineering: ISRO CSE 2020 | Question: 70



In a class definition with 10 methods, to make the class maximally cohesive, number of direct and indirect connections required among the methods are

- A. 90,0
- B. 45,0
- C. 10,10
- D. 45,45

isro-2020 is&software-engineering normal

Answer key

29.3.23 Is&software Engineering: ISRO CSE 2023 | Question: 60



HIPO chart is also known as

- A. UML diagram
- B. Structure chart
- C. DFD
- D. None of the above

isro-cse-2023 is&software-engineering

Answer key

29.4 Out of Gatecse Syllabus (1)

29.4.1 Out of Gatecse Syllabus: GATE CSE 2014 Set 3 | Question: 19, ISRO2017-78



In the context of modular software design, which one of the following combinations is desirable?

- A. High cohesion and high coupling
- B. High cohesion and low coupling
- C. Low cohesion and high coupling
- D. Low cohesion and low coupling

gatecse-2014-set3 is&software-engineering easy isro2017 out-of-gatecse-syllabus

Answer key

29.5 Project Cost (1)

29.5.1 Project Cost: GATE CSE 2011 | Question: 5, ISRO2016-64



A company needs to develop a strategy for software product development for which it has a choice of two programming languages L1 and L2. The number lines of code (LOC) developed using L2 is estimated to be twice of the LOC developed with L1. The product will have to be maintained for five years. Various parameters for the company are given in the table below.

Parameter	Language L1	Language L2
Man years needed for development	LOC/10000	LOC/10000
Development Cost per man year	₹ 10, 00, 000	₹ 7, 50, 000
Maintenance Time	5 years	5 years

Cost of maintenance per year	₹ 1,00,000	₹ 50, 000
------------------------------	------------	-----------

Total cost of the project includes cost of development and maintenance. What is the LOC for L1 for which of the cost of the project using L1 is equal to the cost of the project using L2?

- A. 4000 B. 5000 C. 4333 D. 4667

gatecse-2011 is&software-engineering project-cost normal isro2016

Answer key 

29.6 Software Coupling (1)

29.6.1 Software Coupling: ISRO-2013-67



Which of the following types of coupling has the weakest coupling?

- A. Pathological coupling B. Control coupling
C. Data coupling D. Message coupling

isro2013 is&software-engineering software-coupling non-gatecse

Answer key 

29.7 Software Metrics (1)

29.7.1 Software Metrics: ISRO CSE 2018 | Question: 58



The Functions Point (FP) metric is

- A. Calculated from user requirement B. Calculated from lines of code
C. Calculated from software complexity assessment D. None of the above

isro2018 software-metrics is&software-engineering non-gatecse

Answer key 

29.8 Software Productivity (1)

29.8.1 Software Productivity: ISRO2015-60



If in a software project the number of user input, user output, enquiries, files and external interfaces are (15, 50, 24, 12, 8), respectively, with complexity average weighing factor. The productivity if effort = 70 percent-month is

- A. 110.54 B. 408.74 C. 304.78 D. 220.14

isro2015 is&software-engineering software-productivity non-gatecse

Answer key 

29.9 Software Reliability (1)

29.9.1 Software Reliability: ISRO CSE 2007 | Question: 79



Which of the following models used for software reliability

- A. Waterfall B. Musa C. COCOMO D. Rayleigh

isro2007 is&software-engineering software-reliability

Answer key 

29.10 Software Testing (12)

29.10.1 Software Testing: ISRO CSE 2007 | Question: 67



To execute all loops at their boundaries and within their operational bounds is an example of

- A. Black Box Testing
- B. Alpha Testing
- C. Recovery Testing
- D. White Box Testing

isro2007 is&software-engineering software-testing

Answer key

29.10.2 Software Testing: ISRO CSE 2009 | Question: 67



Activities which ensure that the software that has been built, is traceable to customer requirement is covered as part of

- A. Verification
- B. Validation
- C. Maintenance
- D. Modeling

isro2009 is&software-engineering software-testing

Answer key

29.10.3 Software Testing: ISRO CSE 2009 | Question: 68



A testing method which is normally used as the acceptance test for a software system, is

- A. Regression Testing
- B. Integration Testing
- C. Unit Testing
- D. System Testing

isro2009 is&software-engineering software-testing

Answer key

29.10.4 Software Testing: ISRO CSE 2011 | Question: 34



Black Box Software Testing method focuses on the

- A. Boundary condition of the software
- B. Control structure of the software
- C. Functional requirement of the software
- D. Independent paths of the software

isro2011 non-gatecse software-testing is&software-engineering

Answer key

29.10.5 Software Testing: ISRO CSE 2014 | Question: 23



The test suite (set of test input) used to perform unit testing on a module could cover 70% of the code. What is the reliability of the module if the probability of success is 0.95 during testing?

- A. 0.665 to 0.95
- B. At the most 0.665
- C. At the most 0.95
- D. At least 0.665

isro2014 is&software-engineering software-testing

Answer key

29.10.6 Software Testing: ISRO CSE 2018 | Question: 48



In unit testing of a module, it is found that for a set of test data, at the maximum 90% of the code alone were tested with the probability of success 0.9. The reliability of module is

- a. Greater than 0.9
- b. Equal to 0.9
- c. At most 0.81
- d. At least 0.81

isro2018 is&software-engineering software-testing

Answer key

29.10.7 Software Testing: ISRO CSE 2023 | Question: 39



The MCDC software testing technique stands for

- A. Multiple Conditions Decision Coverage
- B. Multiple coverage Decision conditions
- C. Modified Condition Decision Coverage
- D. None of the above

isro-cse-2023 non-gatecse software-testing

Answer key

29.10.8 Software Testing: ISRO CSE 2023 | Question: 40



Regression testing focuses on validating

- A. module structure
- B. module interfaces
- C. software changes
- D. None of the above

isro-cse-2023 is&software-engineering software-testing

Answer key

29.10.9 Software Testing: ISRO-2013-68



Which of the following testing methods uses fault simulation technique?

- A. Unit testing
- B. Beta testing
- C. Stress testing
- D. Mutation testing

isro2013 is&software-engineering software-testing non-gatecse

Answer key

29.10.10 Software Testing: ISRO-2013-69, UGCNET-Dec2012-III: 62



If a program P calls two subprograms $P1$ and $P2$ and $P1$ can fail 50% of the time and $P2$ can fail 40% of the time, what is the failure rate of program P ?

- A. 50%
- B. 60%
- C. 70%
- D. 10%

isro2013 is&software-engineering software-testing non-gatecse ugcnetcse-dec2012-paper3

Answer key

29.10.11 Software Testing: ISRO-DEC2017-77



Consider the following :

A. Condition Coverage	p. Black box testing
B. Equivalence Class partitioning	q. System testing
C. Volume Testing	r. White box testing
D. Beta Testing	s. Performance testing

Matching A, B, C, D in the same order gives,

- A. r,p,s,q
- B. p,r,q,s
- C. s,r,q,p
- D. q,r,s,p

isrodec2017 software-testing is&software-engineering

Answer key

29.10.12 Software Testing: ISRO2015-59



Alpha and Beta testing are forms of

- A. Acceptance testing
- B. Integration testing
- C. System testing
- D. Unit testing

isro2015 is&software-engineering software-testing

Answer key

29.11

Spiral Model (1)

29.11.1 Spiral Model: GATE IT 2007 | Question: 4, ISRO2016-66



In the Spiral model of software development, the primary determinant in selecting activities in each iteration is

- A. Iteration size
- B. Cost
- C. Adopted process such as Rational Unified Process or Extreme Programming
- D. Risk

gateit-2007 is&software-engineering spiral-model normal isro2016

Answer key

29.12

Uml (3)

29.12.1 Uml: ISRO CSE 2007 | Question: 65



Which of the following is not a UML DIAGRAM?

- A. Use Case
- B. Class Diagram
- C. Analysis Diagram
- D. Swimlane Diagram

isro2007 is&software-engineering uml

Answer key

29.12.2 Uml: ISRO CSE 2007 | Question: 78



In UML diagram of a class

- A. state of object cannot be represented
- B. state is irrelevant
- C. state is represented as an attribute
- D. state is represented as a result of an operation

isro2007 is&software-engineering uml

Answer key

29.12.3 Uml: ISRO CSE 2011 | Question: 28



The above figure represents which one of the following UML diagram for a single send session of an online chat system?

- A. Package diagram
- B. Activity diagram
- C. Class diagram
- D. Sequence diagram

isro2011 is&software-engineering uml

Answer Keys

29.1.1	A
29.2.5	C
29.3.4	B
29.3.9	B
29.3.14	C
29.3.19	D
29.4.1	B
29.9.1	D
29.10.5	B
29.10.10	C
29.12.2	C

29.2.1	A
29.2.6	C
29.3.5	A
29.3.10	D
29.3.15	D
29.3.20	B
29.5.1	B
29.10.1	D
29.10.6	C
29.10.11	A
29.12.3	B

29.2.2	B
29.3.1	C
29.3.6	B
29.3.11	B
29.3.16	C
29.3.21	D
29.6.1	D
29.10.2	B
29.10.7	C
29.10.12	C

29.2.3	C
29.3.2	A
29.3.7	B
29.3.12	A
29.3.17	B
29.3.22	B
29.7.1	C
29.10.3	D
29.10.8	C
29.11.1	D

29.2.4	C
29.3.3	D
29.3.8	A
29.3.13	D
29.3.18	A
29.3.23	B
29.8.1	B
29.10.4	C
29.10.9	D
29.12.1	C